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Melting behavior of nanometer sized gold isomers

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Abstract

In the present work, the melting behavior of nanometer sized gold isomers was studied using a tight-binding potential with a second momentum approximation. The cases of cuboctahedra, icosahedra, Bagley decahedra, Marks decahedra and star-like decahedra were considered. We calculated the temperature dependence of the total energy and volume during melting and the melting point for different types and sizes of clusters. In addition, the structural evolutions of the nanosized clusters during the melting transition were monitored and revealed. It is found that the melting process has three characteristic time periods for the intermediate nanosized clusters. The whole process includes surface disordering and reordering, followed by surface melting and a final rapid overall melting. This is a new observation, which it is in contrast with previous reports where surface melting is the dominant step. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Gold; Surface melting; Computer simulations; Clusters

1. Introduction

The structures and structural evolutions, as well as their related behavior and properties, have fascinated scientist for many years. The richness of the diverse structures of gold clusters, observed experimentally, has posed an obvious challenge and considerable efforts over two decades. Much progress has been made after the unexpected discovery of non-crystallographic structure types (multiply twinned particles MTPs: icosahedra and the so-called Ino and Marks decahedra) [1], or structural transitions (shape fluctuations [2], and melting [3–7]).

As far as structural stability and thermodynamics are concerned, Whetten and coworkers have done many researches on gold clusters passivated by alkythiolates [8,9]. In their experiments, the clusters selectively formed specific sizes, which were isolated by fractional crystallization. Each fraction had a narrow size distribution. The clusters presumably formed these specific sizes because of

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their stability. Furthermore, it is thought that the passivating surface layer does not perturb the structure significantly. If this is true, then the observed structure certainly reflects those of the free clusters.

Consequently, most of the formed cluster sizes were close to those expected for truncated octahedra as has been proved. The comparison between experimental X-ray diffraction patterns and those calculated from structural models led to the identification of the fractions with $N \sim 75$, 101 and 146 as Marks decahedra [10] and those with $N \sim 225$, 459 as twinned truncated octahedron (N denotes the number of atoms). Subsequently, a fraction with $N \sim 38$ was also isolated, the diffraction patterns suggested that it was the truncated octahedron.

Cleveland et al. performed a number of theoretical studies to interpret the experimental results of passivated gold cluster [11,12]. A few sequences of structures were examined by embedded-atom method. As a result, the truncated octahedra and Marks decahedra were found most stable.

Doye and Wales [13] showed very good agreement between the Whetten and coworkers experimental data and their simulated results from Sutton–Chen potential. The two clusters observed experimentally for $N \leqslant 80$ were the two that they found to be the most stable, namely, the 38-atom truncated octahedron and 75-atom Marks decahedron. Garzon et al. studies [14] showed that the lowest energy structures of bare gold cluster with $N = 38$, 55, 75 are disordered. In principle, empirical many-body potential is questionable for such a small cluster. Therefore, the latter results are more reliable while the apparent inconsistence with experimental observation is still worth being studied; of course, the cluster structure is not uniquely determined by energy minimization.

Erocolessi [3] studied the dynamics of small gold particles of $N = 100$–900 with the type of cuboctahedra, surface melting was for the first time proposed. Subsequently, Lewis et al. [4] performed some simulations on the gold clusters of $N = 1055$, 767 for the cuboctahedra type. The results confirmed the conclusion drawn by Erocolessi et al. and further revealed the behavior of coalescence between two clusters. Experimentally, high-resolution transmission electron microscopy (HRTEM) analysis of nanostructures has shown the possibility of directly observing changes in the structure and much work has been done [15–18], in which the electron beam irradiation is used as a stimulation source. However, it is necessary to have sufficient understanding on the critical energy to induce changes of the nanoparticles, these parameters must be related to the size and shape of nanostructures [19].

In our simulations, the cluster sizes cover a wide range from tens of atoms to over twenty thousands atoms. However, more attention has been paid on the relatively large size range, considering that a lot of research work has been done at the range of less than 200 atoms. An exhaustive search for the minimum energy structure in this size range is out of question, owing to exponential increase in the number of nearly equivalent local energy minimum. This problem becomes acute in the present case, owing to the well-known general tendency of gold surface contract and produce more closed package. In the work, several typical isomers of gold usually observed in experiment, which are cuboctahedra, icosahedra, decahedra, Marks decahedra, Marks decahedra with deep truncations models, were constructed and studied (Marks decahedra with deep truncations here have star-like shape, so we also call it star-like decahedra). In most experimental cases, it is a mixture of different structures at any given size, which represents statistical thermodynamics of the energies of different structures plus the effect of kinetics during the growth process. Additionally, small clusters and large clusters may behave somewhat differently both on thermodynamic stability and on the melting behavior. This must lead to new results; it has also raised questions about how structures in small particles should be considered.

In our study, the following factors are highlighted: intermediate size, multitwinned structure, and thermodynamic properties as well as dynamic behavior. We think that our work might be of help to clarify fundamental phenomena for the research in thin film growth, cluster-assembling materials and other fundamental studies for the development for nanotechnology.

2. Computational methods

It is generally recognized that empirical many-body potential have become the main, reliable method in modeling relatively large system usually containing hundreds, thousands, even tens of thousands atoms. Actually, many-body potentials, typically EAM, *N*-body potentials, TB-SMA, Glue model, have been extendedly applied in nearly all fields related to atom configuration. In the case of gold, as mentioned above, both experiment and simulation have received much attention. Therefore, each typical many-body potential model mentioned were developed and applied in the studies of cluster stability, surface reconstruction and many other fields. Each gold potential can reproduce with good accuracy some thermodynamic and dynamic properties. This is mainly because of the fitted experimental values with relatively high accuracy, as well as half full electronic structure for gold, which allows simple mathematics form to describe the complicated interaction with a high accuracy. On the other hand, there are some unavoidable disadvantages, most of which results from the fundamental approximation of the current many-body models.

In the present work, we choose tight-binding potential with a second-momentum approximation (TB-SMA) as proposed by Tomanek et al. [20], which has the following form:

$$E_c = \sum_{i=1}^{N} \left[\sum_{j=1}^{N} A e^{-p(r_{ij}/r_0-1)} - \left[\sum_{j=1}^{N} \xi^2 e^{-2q(r_{ij}/r_0-1)} \right]^{1/2} \right] \quad (1)$$

where the parameters have been determined through fitting the experimental values of cohesive energy and elastic constants, and the structural stability of the ordered phase at ground state (see in detail in Ref. [21]). The potential has been smoothly truncated up to the fifth neighbors. As for gold, the agreement reached on the fitted quantities is excellent. The values of point defect properties calculated with the model are in reasonable accordance with experiments. In addition, it has ever been calculated the pair correlation function of liquid and surface energies of liquid, all the results agree satisfactorily with experiments [22]. However, this model for gold fails to give phonon properties with reasonable accuracy. Generally speaking, we still believe that this model can reproduce the cluster stability and related behavior and properties wit some certain accuracy. Additionally, this model composed of exponent functions, which results in high numeric accuracy and numeric reproducibility.

The MD simulations were performed using a XMD developed by the group of Riffkin, at the Center for Simulation, University of Connecticut [23]. The program employs a predictor–corrector algorithm to integrate the equation of motion. In all the simulations reported here, a time step of 5×10^{-15} s was used. It is necessary to ensure the accuracy of energy calculation due to small energy difference among different isomers of gold, therefore, long runs of usually 2×10^5 time steps are adopted in order to obtain fully relaxed configurations, after that the equilibrated nanoclusters were heated at the heating rate of 5×10^{12} K/s. This heating rate is too large from the point of view of experiment, but it will not cause a large amount of overheating in computer simulations as proven by previous work [24,25].

In this study, several typical structures of gold nanoclusters sequences are studied. The first sequence is cuboctahedra with the magic number from 13 to 22,473 atoms. The second sequence is decahedra with the magic number from 23 to 22,505 atoms. The third sequence is icosahedra, which just has the same distribution of the number of atoms with cuboctahedra sequence. The fourth sequence is Marks decahedra with the magic number from 49 to 22,301 atoms. The last sequence is star-like decahedra with the magic number from 18 to 22,473 atoms. For relatively large clusters including about 25,000 atoms, the corresponding gyration radii for icosahedron and star-like decahedron are 1.8 and 6.2 nm, respectively; cuboctahedron, decahedra, and Marks decahedron have increasing gyration radii at the range from 1.8 to 6.2 nm. The schematic diagram of star-like decahedron, Marks decahedron and regular decahedron are shown in Fig. 1.

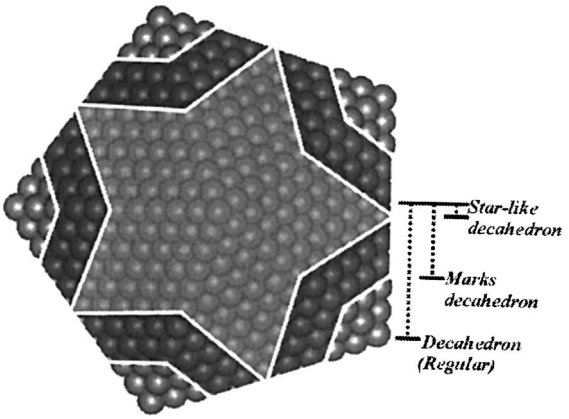

Fig. 1. Structural schematic diagram of star-like decahedron, Marks decahedron and regular decahedron.

3. Results and discussion

3.1. The dependence of the total energy and volume on temperature during heating

In Fig. 2, the dependence of the total energy and volume on temperature during heating processing is shown for the clusters mentioned above. From the figure, the following fact is noteworthy. One is the abrupt increase in all the curves, which indicates the solid–liquid transition. However, unlike the bulk of pure metal, melting most commonly occur at some certain zone rather than a temperature point for most clusters as shown in Fig. 2. It is well known that if the heating rate is too high, partial melting processing will occur. The curve of total energy will behave in the similar way. Nevertheless, the heating rate 5×10^{12} K/s is not large enough to result in overheating or partial melting processing for computer simulations. In this case, that apparent melting zones occur, there must exist another reason, the subsequent results demonstrate that surface disordering and reordering as well as surface melting lead to that fact. According to their melting behavior, the clusters can be divided into two different groups: for cuboctahedra, decahedra, and icosahedra, in the whole size range under study, surface melting is considerable and no abrupt increases like that of the bulk of pure metal are observed. For Marks decahedra and

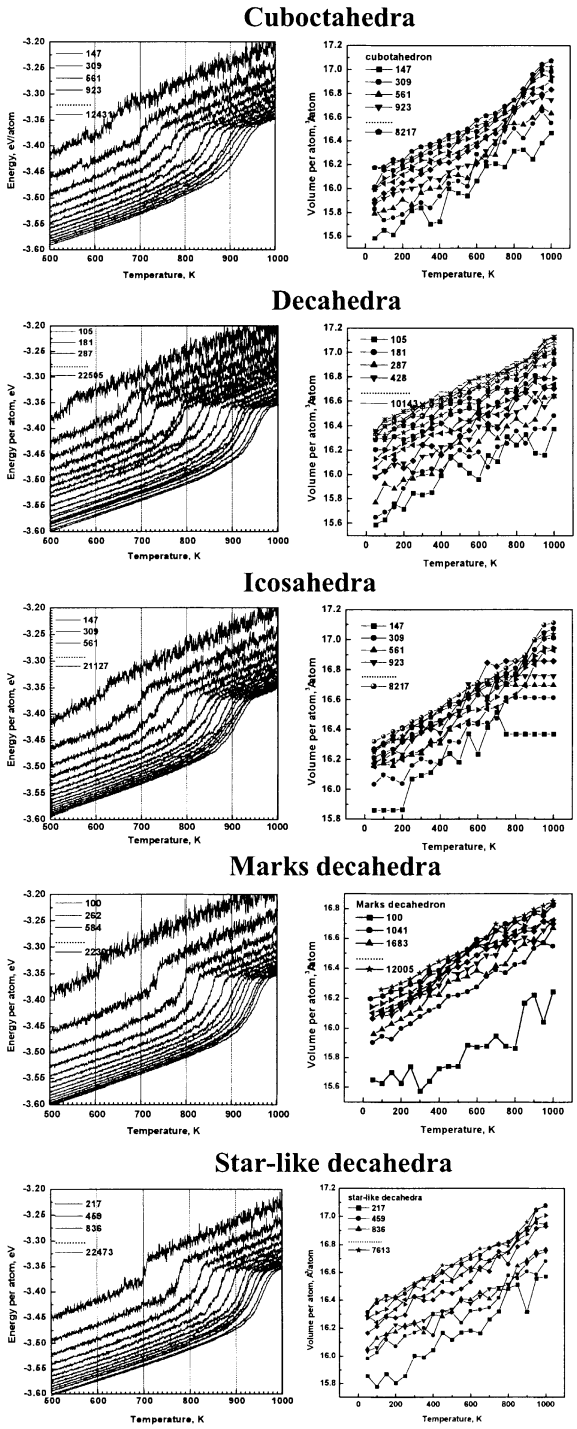

Fig. 2. Total energy and volume dependence of different types of clusters to temperature. Cuboctahedra, decahedra, icosahedra, Marks decahedra and star-like decahedra configurations are considered.

star-like decahedra with less than about 1000 atoms, there is an abrupt increase like the behavior of bulk pure metal, which indicates that surface melting is not considerable, so in comparison with the other cluster structures with almost identical sizes; they occupy relatively high melting points. However, beyond 1000-atom clusters and most conspicuous beyond 10,000-atom clusters, surface melting becomes nearly the same with those of the types of clusters. The second noteworthy point is height of melting zone, which reflects the amplitude of melting latent heat. From Fig. 2, it can be seen that the melting latent heat consistently increases from small size to the larger one of all type structures, but over 5000 atoms, the melting latent heat approach some certain value. Considering of small difference of melting latent heat among different types of structures and the difficulty of evaluate it accurately because of melting zone, the comparison of melting latent heat among different types of structures was not performed. The third noteworthy point in energy curve is that the slope before melting transition, namely, constant pressure heat capacity are almost independent of the type and size of isomers, their values distribute within 31 ± 0.5 J/mol, which is close to that of the bulk fcc gold, 25.42 J/mol [26]. Volume changes with temperature were also calculated by free volume module of Cerius^2 4.0 [27]. It is notable that for a small size of clusters the volumes take on large fluctuations making difficult to distinguish the indication of solid–liquid transition from that. These clusters cover no larger than 309 atoms of cuboctahedra, 423 atoms of decahedra, 1415 atoms of icosahedra, 1041 atoms of Marks decahedra, 217 atoms of star-like decahedra. In addition to dramatic fluctuations, volume change behaves quite different of that of the bulk as shown in the figure. We think that this is result of two opposite effects: one of them is the thermal expansion, which causes the volume increase, the other must be the morphological evolution, namely, there is a strong tendency of being increasingly rounded, which will give rise to the decrease of the volume. The competition of these two factors must result in complicated behavior of volume change. Volume keep unchanged at low and high temperature region for smaller icosahedra, and at high temperature region for smaller Marks decahedra and even drops with increasing temperature at some certain temperature range for some types of clusters. However for relatively large clusters, the volume change behaves like the bulk, which is because most atoms are constrained by the matrix structure when the clusters are sufficiently large, to say, only those atoms at vertices and surface short of constraint can move and even flow, therefore the morphologies only can change little by little with increasing temperature. The irregularity of volume change, for free clusters are applicable in some fields of functional materials, but the size and temperature range are limited.

3.2. Melting point for different types and size of clusters

From Fig. 2, the melting points are extracted by estimating the temperature of middle points at melting zone. All the melting points for different sizes and types of clusters are displayed in Fig. 3. The inset shows a zoomed-in view for small clusters with no more than 2000 atoms. As shown in Fig. 3, for cuboctahedra with 147, 309, 561 and 923 atoms, the melting point is, respectively, about 645, 706, 734, and 771 K. Experimentally, the melt-

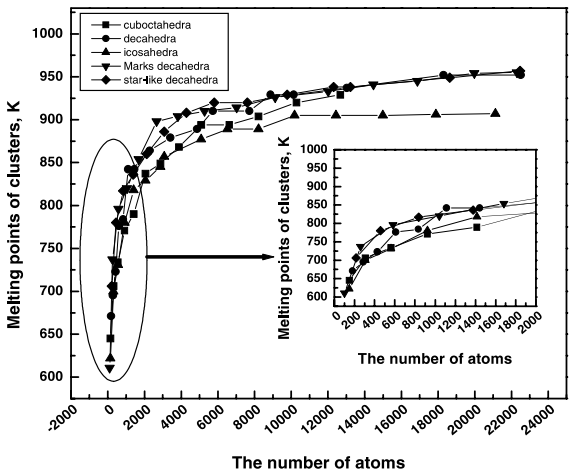

Fig. 3. Melting points versus the sizes of clusters for cuboctahedra, decahedra, icosahedra, Marks decahedra and star-like decahedra.

ing points of several sizes of cuboctahedra clusters are available. Considering the size dispersion in these experiments, an accurate determination of melting points for a specific size is not possible, especially for $d < 2.5$ nm [28], therefore ours simulated results can be compared with those experimental values even if the sizes do not correspond exactly. Our simulated results are generally consistent with the experiments as well as the simulated results from Ercolessi et al. [3], and are significantly lower than those predicted by Pawlow's theory (850, 970 and 1040 K, using parameters from Ref. [28]). This demonstrates that the interatomic interaction potential used in our simulation is reliable. Within the size range of 12,000 atoms, different types of clusters have different melting points, the largest difference of melting points are up to 75 K. Cuboctahedra and icosahedra have the relatively low melting point and Marks decahedra and star-like decahedra have relatively high melting points, and the melting points of regular decahedra locate at intermediate region and exist relatively large fluctuations. Beyond the size as large as 12,000 atoms, all types of the clusters except for icosahedra have almost the same melting points, and increase slowly. From the tendency of the curve, the melting points of the clusters still have value far from that of the bulk of gold, namely 1337.3 K [26]. This demonstrates that the existence of surface melting enormously drops melting points. We can infer that it cannot approach melting point of the bulk until a considerable larger size from the point of view of atom scale. The mere exception, icosahedra, almost keeps the melting points unchanged beyond the size as large as 10,000 atoms. Based on results on liquid metals [24,25], it is suggested that liquid gold mainly compose of icosahedral order, in other words, many icosahedra distribute in the whole system connectedly or separately. Possibly, when the size of system is sufficiently large, this kind of distributions can be built. As a result, the size does not influence largely the statistical structure of liquid and hence the melting points. On the contrary, as the size is not sufficiently large, the statistical structure the liquid must vary with the size and giving rise to the different melting points.

3.3. Structural evolution during melting transition

For the sake of a clear view of structural evolution, in particular surface atoms, we introduce pair correlation functions, which is the most commonly used in evaluation the structural characteristics of disordered structure. It is also the only experimentally available physical quantity. For clusters, the center region usually have a little change from the point of view of distance and coordinates of atoms, so the pair correlation functions must directly reflect the state the surface atoms. Two typical sizes for each type of clusters were selected and pair correlation functions were calculated every 10 K about above and below 50 K of the corresponding melting points. These clusters include cuboctahedra with 147 and 2869 atoms, decahedra with 105 and 2289 atoms, icosahedra 147 and 2869 atoms, Marks decahedra with 100 and 2665 atoms, star-like decahedra with 217 and 3077 atoms as shown in Fig. 4. For all types of small clusters, there are many fluctuations, but the indication of solid–liquid transition still can be observed, especially for star-like decahedra. For all types of large clusters in Fig. 4, pair correlation functions clearly reveal fundamental characteristics and the solid–liquid transition. For example, at the lowest temperature in each graph, it can be seen that all types of clusters have their own features, unlike the periodic structure; they usually have widen in the first and second peaks. They look more like those of a disordered structure except for two small peaks between first and second neighbors as well as second and third neighbors (which are usually the feature of disordered structure mixing with some crystallite). Additionally at this size, the split of the second peak on the pair correlation function of the icosahedra is not obvious near the melting point, this is also a difference between icosahedra cluster and a bulk mainly composed of icosahedra.

Trajectories were recorded every 10 K during heating processing. Therefore, the detailed transition processing during heating for all types and sizes of clusters were observed and examined. Fig. 5 shows two typical sizes for each type of structure at some critical temperatures. For cuboctahedron with 147 atoms, only atom vibration can

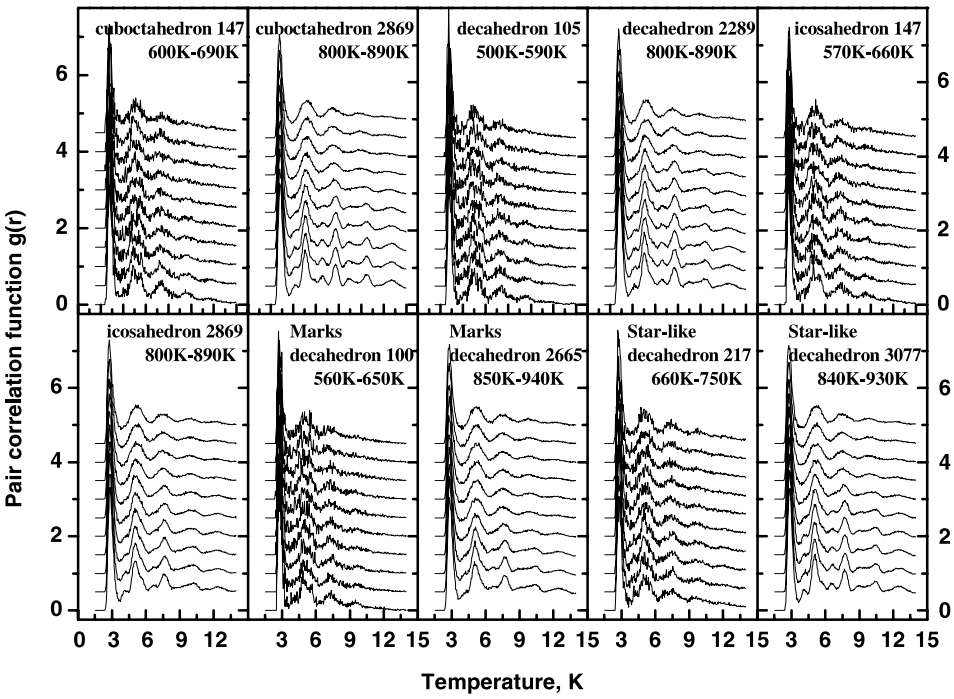

Fig. 4. Pair correlation functions near melting points for ten typical size and structure of clusters. Cuboctahedra ($N = 147$ and $N = 2869$), decahedra ($N = 105$ and $N = 2289$), icosahedra ($N = 147$ and $N = 2869$), Marks decahedra ($N = 100$ and $N = 2665$), and star-like decahedra ($N = 217$ and $N = 3077$).

be identified below 400 K. We also attempt to find out some interesting structure transitions by changing the orientation, as a result, it is found that cuboctahedron–icosahedron transition from 410 to 420 K, which can be observed easily in the orientation $[0, -\sqrt{3}, 3]$. This kind of transition is the so-called quasimelting [1] and commonly occur in a second scale or longer time scale at the room temperature. However, our results indicate that cuboctahedron–icosahedron transition complete in the order of 10^{-12} s during heating processing. After this transition completes, icosahedra persist about 100 K and then start to melt, the melting processing is similar with the identical size of icosahedron as shown in this figure. On the other hand, the surface melting cannot be obviously observed due to the small size. Similarly, cuboctahedron with 309 atoms also have this kind of cuboctahedron–icosahedron transition. However, when the size is as large as 561 atoms the cuboctahedron–icosahedron can no longer be observed. The cuboctahedron with 2689 atoms behave differently from that with 147 atoms, the obvious melting process is displayed. It can be clearly distinguished that first the vertices start to disorder and then spread to edges and finally proceed inward. Despite of the surface melting there are some fluctuations, sometimes molten atoms in surface can recover the ordering state due to nucleation effect of the matrix still in an ordered state. This fluctuation of surface disordering and reordering holds for a relatively long time. When all vertices and edges disorder, the fluctuation of disordering and reordering cease, and then surface melting goes on and the structure of surface becomes liquid-like. As soon as the surface melting spreads inward, the overall melting completes rapidly. This just explains why the bottom side of abrupt elevation in the curve of total energy spans a wider temperature range than the topside of that. Accompanying the surface disordering and reordering as well as the surface melting,

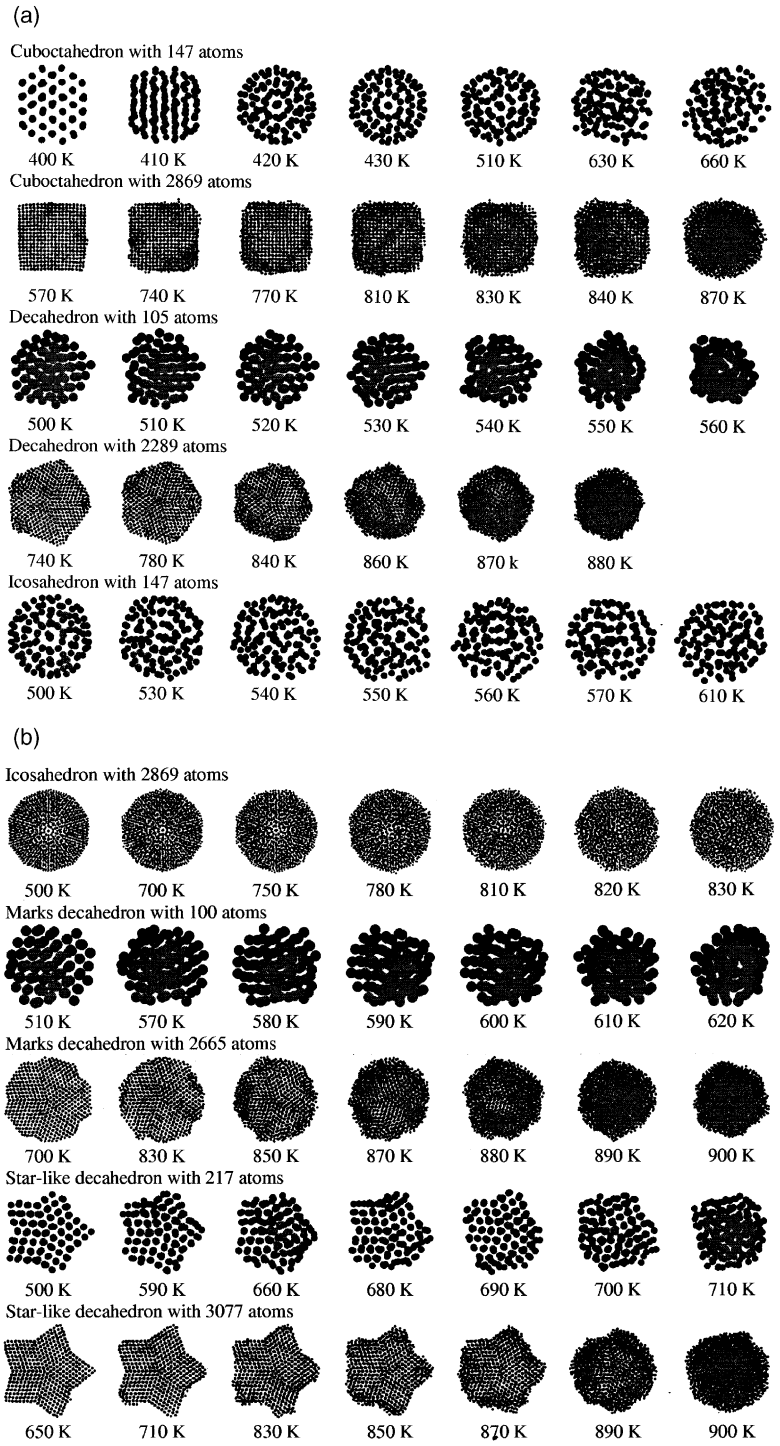

Fig. 5. Configuration evolution during heating processing for the icosahedron with 2869 atoms, Marks decahedron with 100 and 2665 atoms, and star-like decahedron with 217 and 3077 atoms.

the morphologies of clusters become increasingly rounded. In comparison with larger cluster, the small clusters change their morphologies more rapidly as shown in Fig. 5, which demonstrates the law of volume change in an intuitive way. Other types of large clusters behave like the cuboctahedron with 2869 atoms in spite of their own characteristics. In summary, the entire melting processing can be span into three characteristic periods for intermediate nanosized clusters. The first is a long surface disordering and reordering, the next is a short time of surface melting in which surface disorders completely, and finally followed by rapid overall melting.

3.4. Dynamic analysis by mean square displacements

For studying the dynamic process in a quantitative way, the mean square displacements were calculated. We do not take into account the initial configuration as reference, but take the previous configuration in trajectory of heating processing, namely, the configuration at a temperature of 10 K low. In this manner, the mean square displacements more directly reflect the change of diffusivity of atoms at different temperatures.

From Fig. 6, it can be found that the diffusion of atoms in the clusters is considerably small below to some temperature, and then increase almost linearly at a relatively small rate. Small clusters have a more obvious behavior as displayed in Fig. 6. Actually, the large clusters behave almost the same if the details are zoomed in. This confirms the relatively large temperature range of surface disordering and reordering in a quantitative way. After that, a rapid melting transition happens within a narrow temperature zone, followed by relatively large thermal expansion of liquid.

At the right bottom corner of Fig. 6, it can be noticed that the clusters as small as tens of atoms behave quite differently from that of larger clusters from the point of view of mean square displacements. Furthermore, different types of clusters behave differently even if the sizes are almost identical, most of them have sudden increases and keep this tendency until elevated temperature, which is more like the behavior of sublimation rather than melting, but we find that all systems still keep in condensed status. For Marks decahedron with 100 atoms and decahedron with 105 atoms as well as star-like decahedron with 80 atoms, despite obvious indications of solid–liquid transitions, the thermal expansions are tens of times as large as those of the identical types except for relatively large size of clusters. It is surprising to find that the mean square displacements of cuboctahedron with 13 atoms look like plane waves. As well known, for quite small systems commonly with several atoms or tens of atoms, the empirical many-body potentials as well as classical molecular dynamics simulation technique are questionable and face with a large challenge, therefore, here these results are only proposed as a reference.

All the above-mentioned clusters are in unsupported state, namely, the interaction between substrate and clusters is not involved. The most common cases in HRTEM are amorphous carbon or silicon, both of which have quite complicated interaction with transition metals, to date no accurate interaction potential is available for this kind of system with any compositions. Fortunately, there are many cases of loose contact between substrate and clusters results in a week interaction; in other case, particles are observed in edges with little interaction with the substrate. Therefore, in some cases computer simulation of free cluster can reproduce experimental phenomena in some sense.

4. Summary and conclusions

The melting processing of intermediate size and all types of clusters clearly include three stages: a relatively long time of surface disordering and reordering, a short time of surface melting and final rapid overall melting. Ercolessi et al. reported surface melting through diffusion coefficients [3], but we think that surface melting alone cannot describe exactly this kind of transition processing. Through our direct observation, quantitative and qualitative analysis, three stages of melting processing including surface melting are found. This process significantly drop the melting point of clusters, even if the cluster is as large as 25,000 atoms,

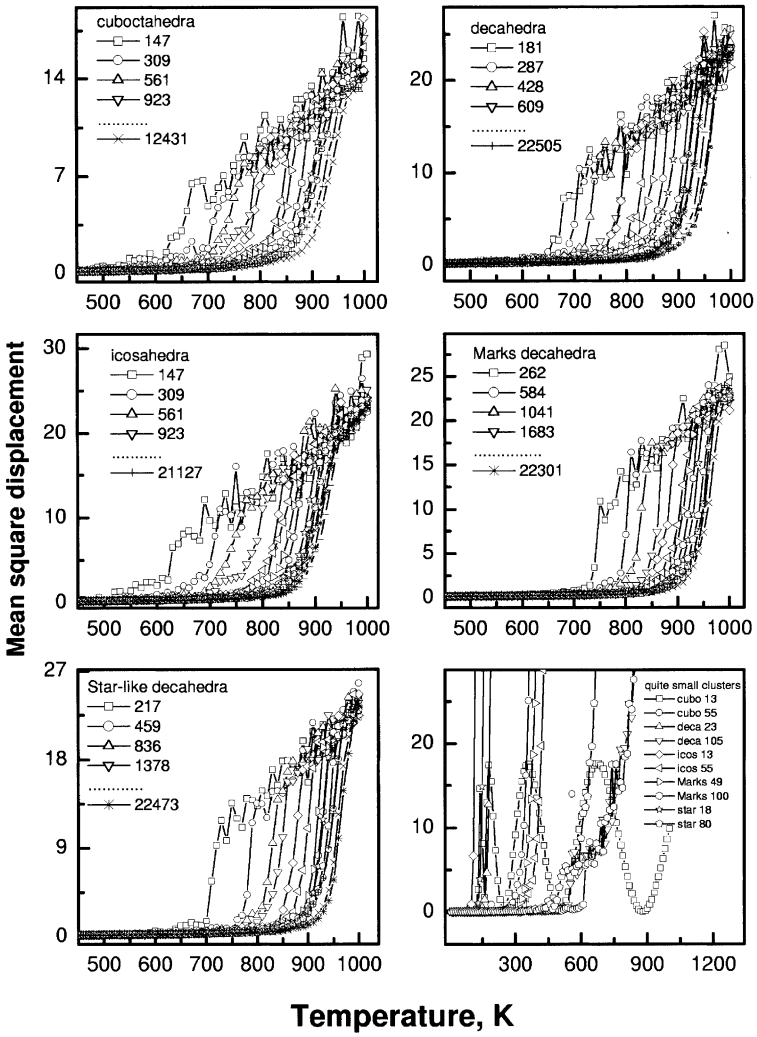

Fig. 6. Mean square displacements obtained every 10 K; the reference configuration is not initial one, but the previous temperature configuration point.

melting points are still far away from that of the bulk. Moreover the melting points of clusters increase quite slowly with increasing sizes in the size range over 25,000 atoms, so we can infer that melting points of cluster cannot approach the equilibrium melting point of the bulk until the size is considerably large (from the point of view of atomic scale). Icosahedra have the lowest melting points and star-like decahedra are the most hard to melt. Melting points of regular decahedra locate middle of them and have large fluctuations at different sizes. In the case of nearly identical size, melting points for different types of clusters may differ by up to 75 K. When the cluster size is over about 12,000 atoms, all types of clusters except for icosahedra behave almost the same from the point of view of melting. Nevertheless, the melting points of icosahedra almost keep unchanged with increasing sizes in that size range. With respect to the melting latent heat, there are no large differences among different types of clusters in the case of similar sizes.

The volume changes of different clusters during heating behave differently, which is mainly resulted from two competitive factors. One is, of course, thermal expansion; the other is evolution of cluster morphologies. The former results in an increase of volume, but the latter give rise to a decrease of volume. Therefore, it is easy to understand that the morphological difference of different isomers must lead respective characteristics, in particular small size of clusters where atoms are short of constraint from the matrix and tend to flow randomly. The volume changes of clusters enormously differ from those of the bulk, especially for small clusters with no more than 1000 atoms.

For cuboctahedra with 147 atoms and 309 atoms, cuboctahedron–icosahedron transition takes place prior to solid–liquid transition at the order of 10^{-12} s, which corresponds the temperature range of 10–20 K during heating.

From structural analysis of pair correlation functions and dynamic analysis of mean square displacement, the melting characteristics of different clusters are further confirmed in a quantitative way.

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